



POWER BI MAJOR CAPSTONE PROJECT
END TO END
POWER BI PROJECT GUIDE

PROJECT TITLE

SUPERSTORE SALES & PROFIT ANALYTICS
DASHBOARD

PROJECT CATEGORY

Sales Analytics
Profit Analytics
Business Intelligence

INSTITUTE NAME

Infinity Code Nexus

TRAINER / GUIDE

Anand Kumar Anamaina

CREATED BY

K . Aishwarya Reddy

TOOLS USED

Power BI Desktop Power Query, DAX & Power BI
Service

SUBMISSION DATE

[09/06/2026]

Prepared as part of the Power BI Masterclass Program to demonstrate skills in
Data Analysis, Data Modeling, DAX Development, Dashboard Design, UI/UX,
Storytelling, and Business Intelligence Reporting.

PROJECT OVERVIEW

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PROJECT OBJECTIVE

Develop a comprehensive Power BI dashboard to analyze sales, profit, customer behavior, product performance, and regional business performance for a retail superstore.

Participants are expected to:

- ✓Analyze sales performance
- ✓Evaluate profitability
- ✓Understand customer purchasing behavior
- ✓Assess product contribution
- ✓Analyze regional performance
- ✓Generate business insights

PROJECT GOAL

Transform retail transactional data into actionable insights that help organizations monitor sales performance, improve profitability, optimize products, and support strategic decision-making.

BUSINESS PROBLEM

Retail businesses generate large volumes of transactional sales data. However, business leaders often struggle to convert raw data into meaningful insights for performance improvement and business growth.

The objective is to identify:

- Which products generate the highest sales?
- Which categories contribute the most profit?
- Which regions drive business growth?
- How discounts impact profitability?
- Which customer segments contribute most to revenue?
- Which products require immediate business attention?

BUSINESS QUESTIONS

Participants must answer:

1. What is total sales revenue?
2. What is total profit?
3. Which products perform best?
4. Which regions generate maximum revenue?
5. Which customer segments contribute most?
6. How do discounts affect profit?
7. What are sales and profit trends over time?

DATASET INFORMATION

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DATASET NAME : Superstore Dataset

DATASET DETAILS : Approximate Records 9,994+

Columns : 19+

EXPECTED ANALYSIS AREAS

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Sales Analytics

Profit Analytics

Customer Analytics

Product Analytics

Regional Analytics

Shipping Analytics

POWER QUERY TRANSFORMATIONS

DATA CLEANING ACTIVITIES

- ✓ Removed Duplicate Records
- ✓ Handled Missing Values
- ✓ Corrected Data Types
- ✓ Validated Sales and Profit Data
- ✓ Standardized Categories

DATA PREPARATION ACTIVITIES

- ✓ Renamed Columns
- ✓ Removed Unnecessary Columns
- ✓ Created Custom Columns
- ✓ Split Date Columns
- ✓ Replaced Values
- ✓ Created Hierarchies

CUSTOM COLUMNS CREATED

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Profit Margin %
Sales Category
Order Year
Order Month
Discount Category
Customer Segment Group

SCREENSHOT:

	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gute	
2	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gute	
3	12-06-2016	16-06-2016	Second Class	DV-13045	Darrin Van Huff	
4	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell	
5	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell	
6	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	
7	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	
8	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	
9	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	
10	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	
11	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	
12	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	
13	15-04-2017	20-04-2017	Standard Class	AA-10480	Andrew Allen	
14	05-12-2016	10-12-2016	Standard Class	IM-15070	Irene Maddox	
15	22-11-2015	26-11-2015	Standard Class	HP-14815	Harold Pawlan	
16	22-11-2015	26-11-2015	Standard Class	HP-14815	Harold Pawlan	
17	11-11-2014	18-11-2014	Standard Class	PK-19075	Pete Kriz	
18	13-05-2014	15-05-2014	Second Class	AG-10270	Alejandro Grove	
19	27-08-2014	01-09-2014	Second Class	ZD-21925	Zuschuss Donatelli	
20	27-08-2014	01-09-2014	Second Class	ZD-21925	Zuschuss Donatelli	
21						

DATA MODELING

Star Schema

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Purpose:

FACT TABLE

Orders_Fact

DIMENSION TABLES

Dim_Customer

Dim_Product

Dim_Region

Dim_Category

Dim_Order_Date

Dim_Segment

RELATIONSHIPS

One-to-Many Relationships

FILTER DIRECTION

Single Direction

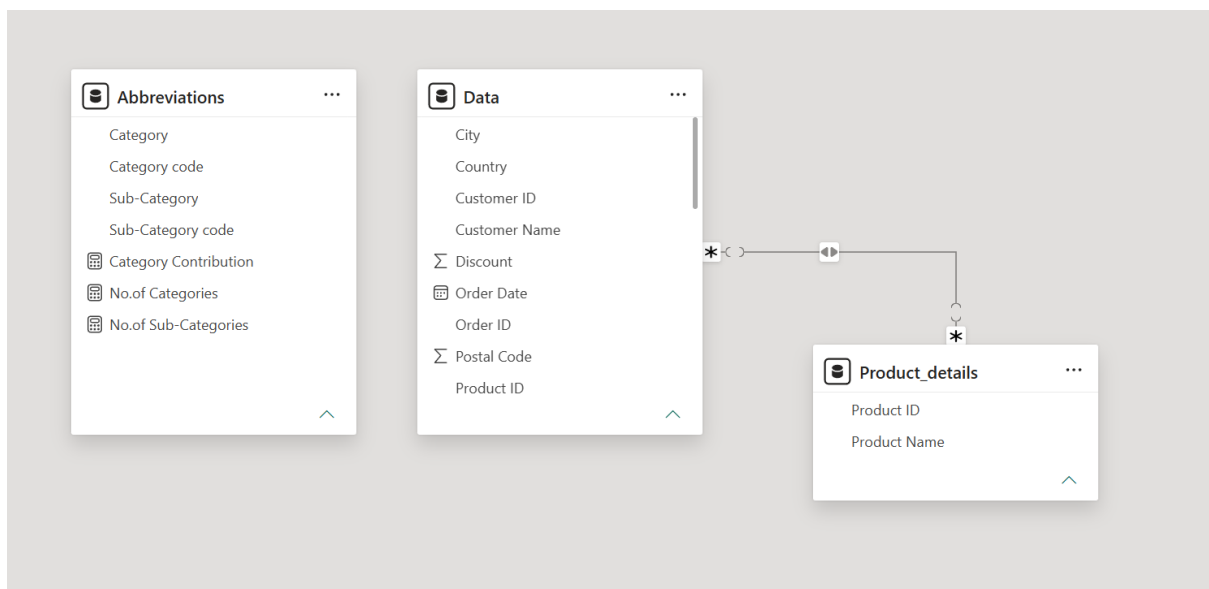
MODEL BENEFITS

Improved Performance

Easy DAX Development

Better Scalability

SCREENSHOT:



DAX MEASURES USED

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TOTAL SALES

SUM(Orders[Sales])

PURPOSE: Calculate total revenue generated.

TOTAL PROFIT

SUM(Orders[Profit])

PURPOSE: Calculate overall business profit.

PROFIT MARGIN %

DIVIDE(SUM(Orders[Profit]),

SUM(Orders[Sales]))

PURPOSE: Calculate profitability percentage.

TOTAL ORDERS

DISTINCTCOUNT(Orders[Order ID])

PURPOSE: Calculate total customer orders.

AVERAGE ORDER VALUE

DIVIDE([Total Sales],[Total Orders])

PURPOSE: Calculate average order revenue.

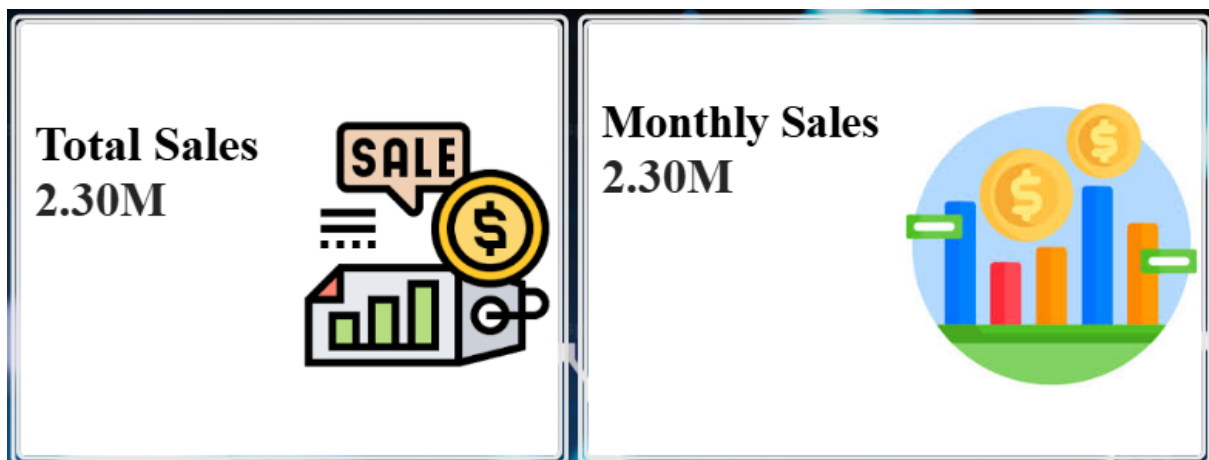
VISUALIZATIONS USED

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KPI CARDS:

Display key business metrics.

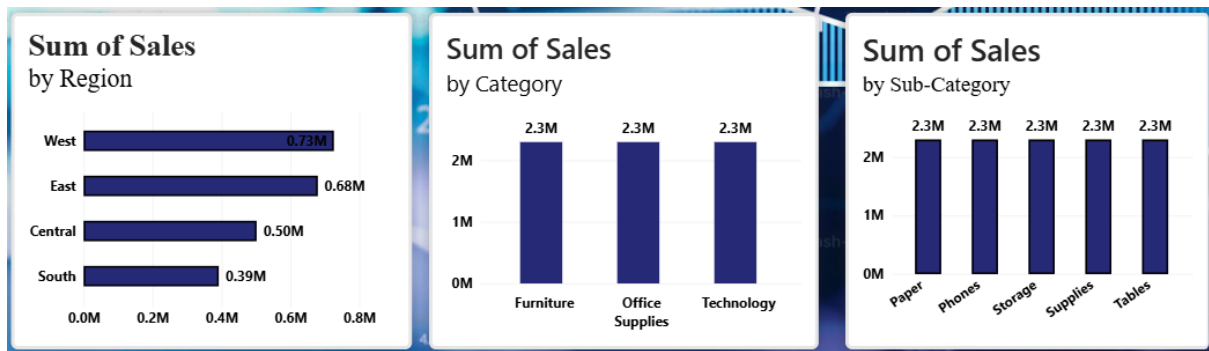
SCREENSHOT:



BAR CHART:

Compare sales and profit across categories and regions.

SCREENSHOT:



LINE CHART:

Show sales and profit trends over time.

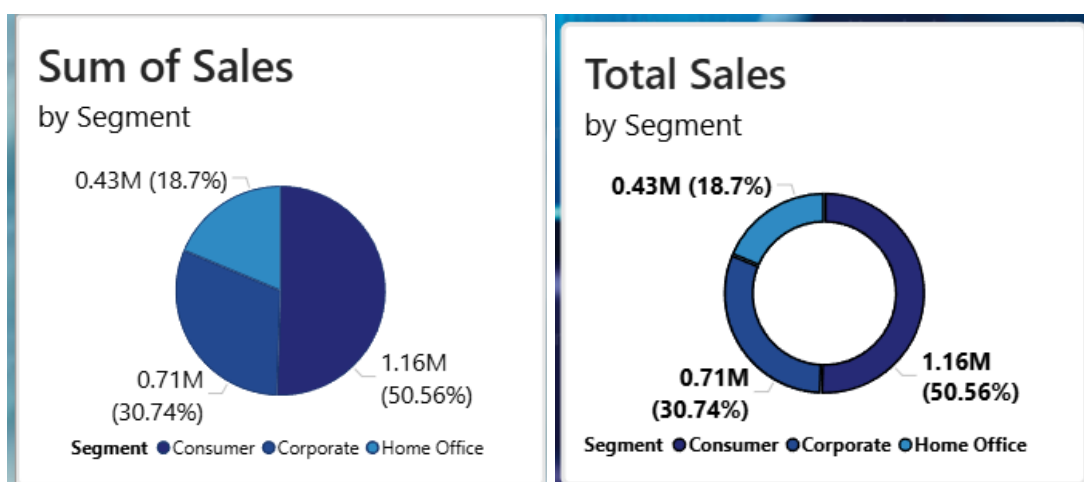
SCREENSHOT:



PIE / DONUT CHART:

Show contribution by category and segment.

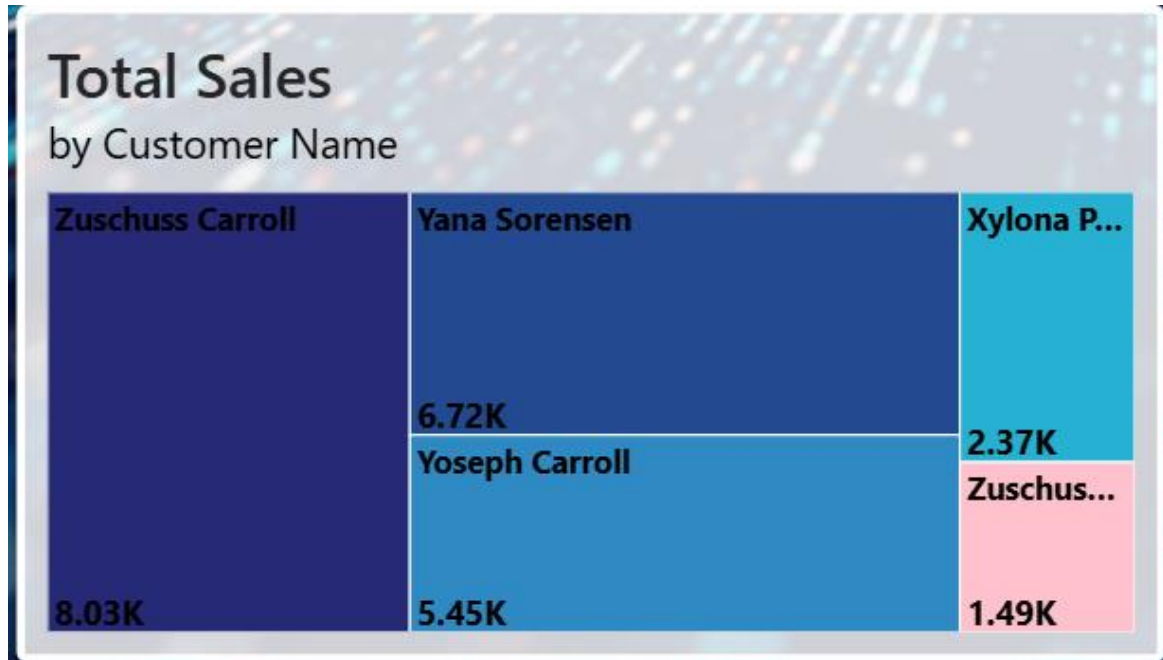
SCREENSHOT:



TREEMAP:

Show category and sub-category contribution.

SCREENSHOT:



MAP VISUAL:

Analyze regional sales performance.

SCREENSHOT:



DASHBOARD SCREENSHOTS

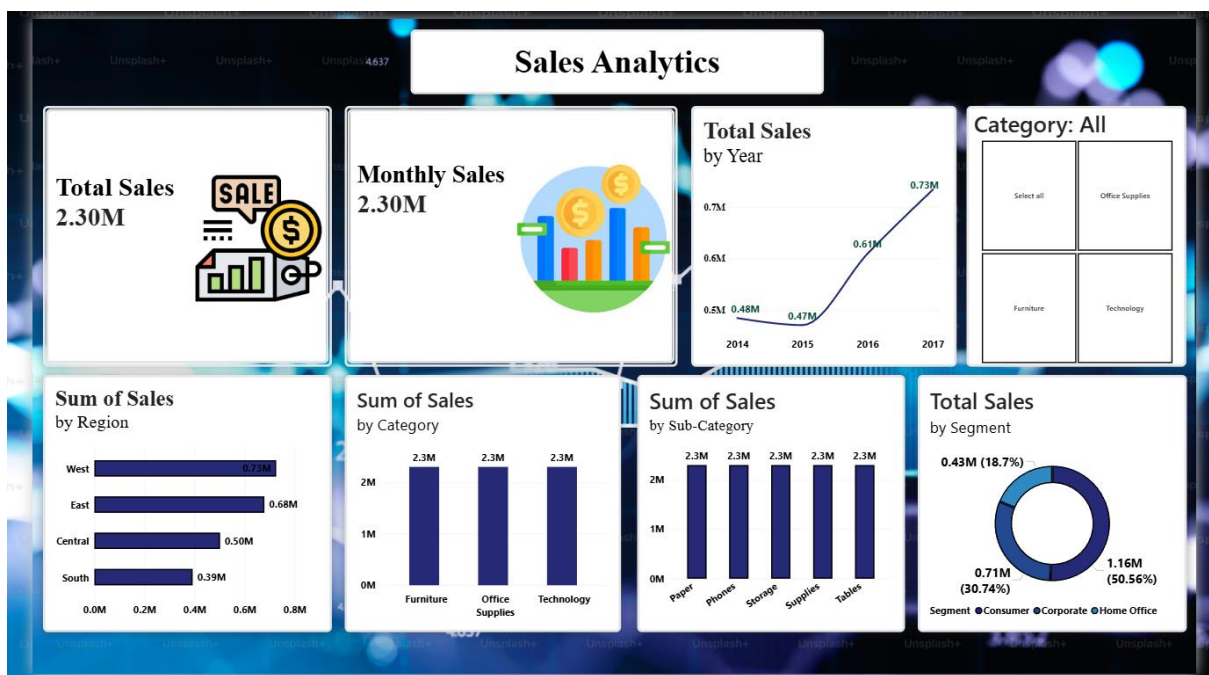
EXECUTIVE OVERVIEW



Explanation :

Provides a high-level summary of sales, profit, KPIs, and overall business performance.

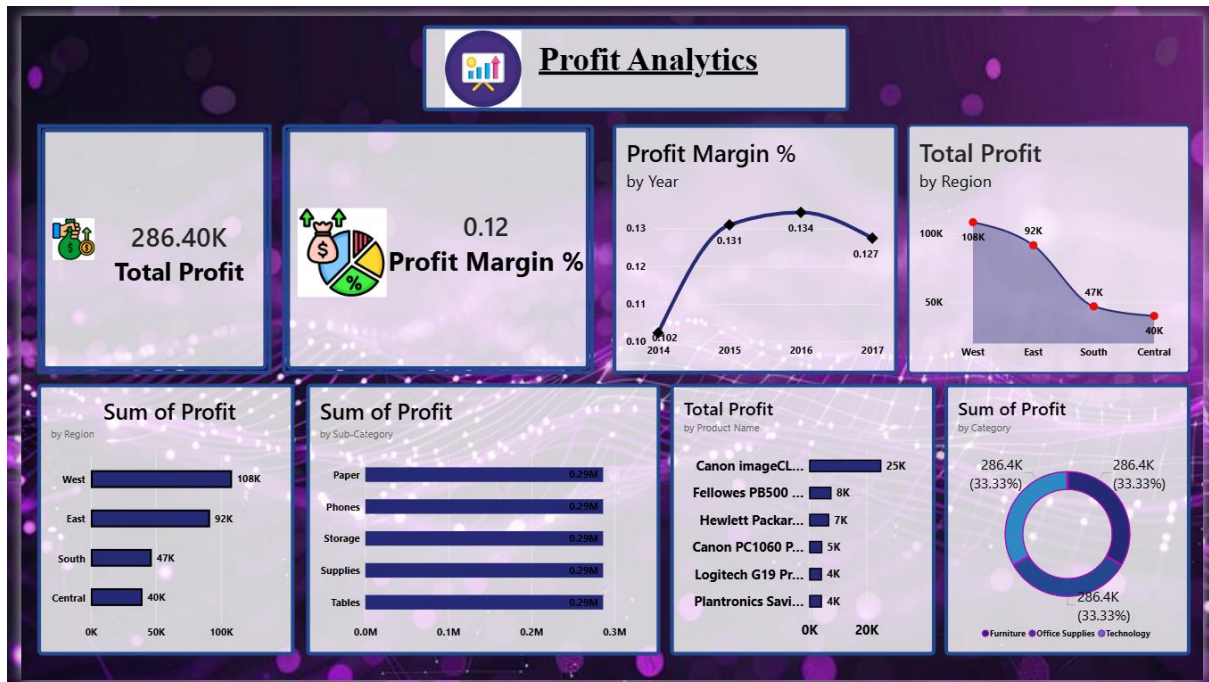
SALES ANALYTICS



Explanation :

Analyzes revenue trends across regions, categories, and customer segments.

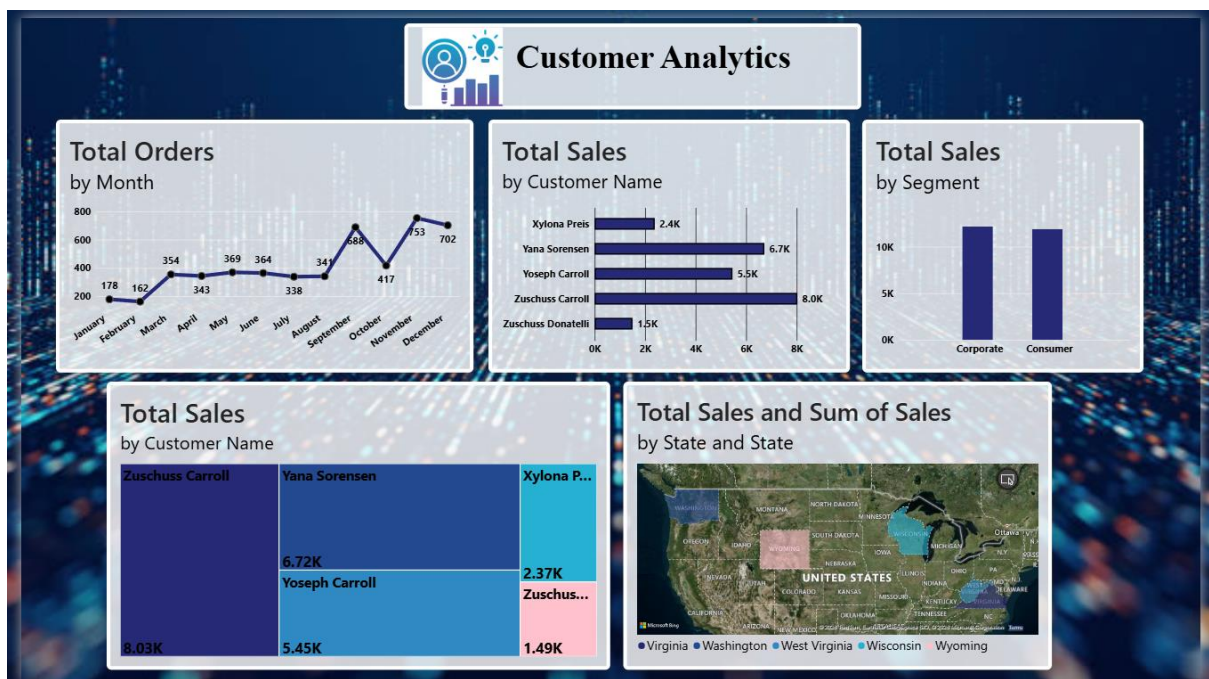
PROFIT ANALYTICS



Explanation :

Analyzes profitability, profit-driving products, and regional profit performance.

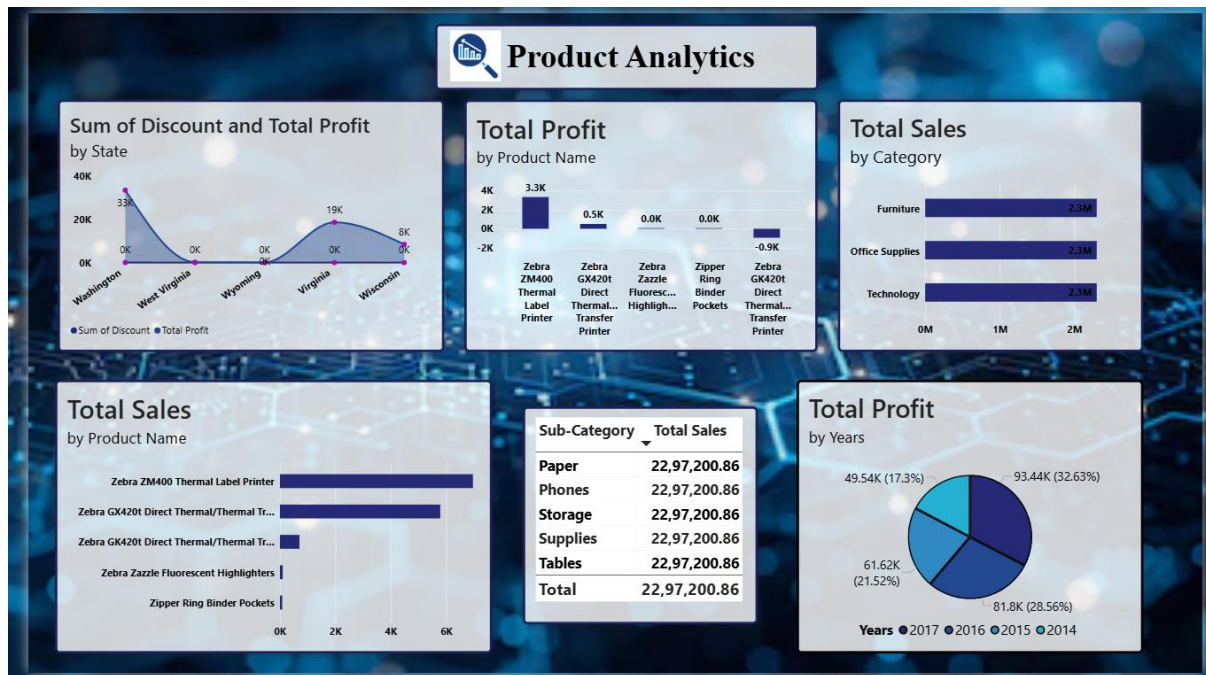
CUSTOMER ANALYTICS



Explanation :

Analyzes customer contribution, segments, and purchasing patterns.

PRODUCT ANALYTICS



Explanation :

Evaluates top-performing products, category contribution, and discount impact.

KEY INSIGHTS

INSIGHT-1

Technology and Office Supplies categories contribute major sales revenue.

INSIGHT-2

Higher discounts negatively affect overall profit margins.

INSIGHT-3

Certain regions consistently generate higher profits than others.

INSIGHT-4

Top customers significantly contribute to business revenue.

BUSINESS RECOMMENDATIONS

RECOMMENDATION-1

Reduce high discount dependency to improve profitability.

RECOMMENDATION-2

Focus marketing efforts on high-performing regions.

RECOMMENDATION-3

Promote profitable products to increase revenue growth.

RECOMMENDATION-4

Strengthen customer retention strategies for repeat purchases.

FEATURES IMPLEMENTED

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- ✓ Drill Down
- ✓ Drill Through
- ✓ Tooltips
- ✓ Dynamic Titles
- ✓ Slicers
- ✓ Conditional Formatting
- ✓ Navigation Buttons
- ✓ Bookmarks
- ✓ Custom Theme

CHALLENGES FACED

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- Data Cleaning Challenges
- Relationship Management
- DAX Optimization
- Dashboard Layout Design
- Performance Tuning

LEARNINGS

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- Power Query
- Data Modeling
- DAX Development
- Dashboard Design
- Storytelling
- Business Analysis

PROJECT STORYTELLING

BUSINESS SCENARIO

Retail organizations generate large volumes of sales and transactional data related to customers, products, discounts, profits, and regions. However, decision-makers often struggle to convert this information into meaningful business insights that support business growth and profitability.

BUSINESS PROBLEM

- Which regions generate maximum revenue and profit?
- Which products drive business growth?
- How do discounts affect profitability?
- Which customer segments contribute more revenue?
- Which categories require business improvement?

ANALYSIS APPROACH

- Analyzed sales and profit across categories and regions.
- Compared product contribution and profitability.
- Evaluated customer purchasing behavior.
- Measured the impact of discounts on profit.
- Identified trends over time for business monitoring.

KEY FINDINGS

Top-performing products generated the majority of sales revenue.
Regions showed different levels of profitability.
Customer segments contributed differently to sales performance.
Higher discounts reduced profit margins significantly.

BUSINESS IMPACT

Supports business decision-making.
Improves profitability planning.
Enhances customer understanding.
Provides data-driven sales insights.
Enables performance monitoring.

PROJECT DEPLOYMENT

DEPLOYMENT OBJECTIVE

Deploy the Power BI solution to enable stakeholders to access interactive dashboards and business insights in real time.

DEPLOYMENT PLATFORM

Power BI Service

DEPLOYMENT PROCESS

STEP-1

Validate Data Model and DAX Measures in Power BI Desktop.

STEP-2

Optimize Dashboard Layout, Performance, and User Experience.

STEP-3

Publish PBIX File to Power BI Service.

STEP-4

Configure Dataset Refresh Settings.

STEP-5

Create Dashboard from Published Reports.

STEP-6

Apply Security and Access Permissions.

STEP-7

Share Dashboard with Business Stakeholders.

BUSINESS BENEFITS

Centralized Reporting Platform
Interactive Sales Analytics
Real-Time Performance Monitoring
Improved Strategic Planning
Faster Data-Driven Decisions

CONCLUSION

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The Superstore Sales & Profit Analytics Dashboard successfully analyzed sales performance, profitability, customer behavior, product contribution, and regional performance.

The project generated valuable insights that can help organizations improve profitability, optimize product strategy, enhance customer engagement, and support better business decision-making.